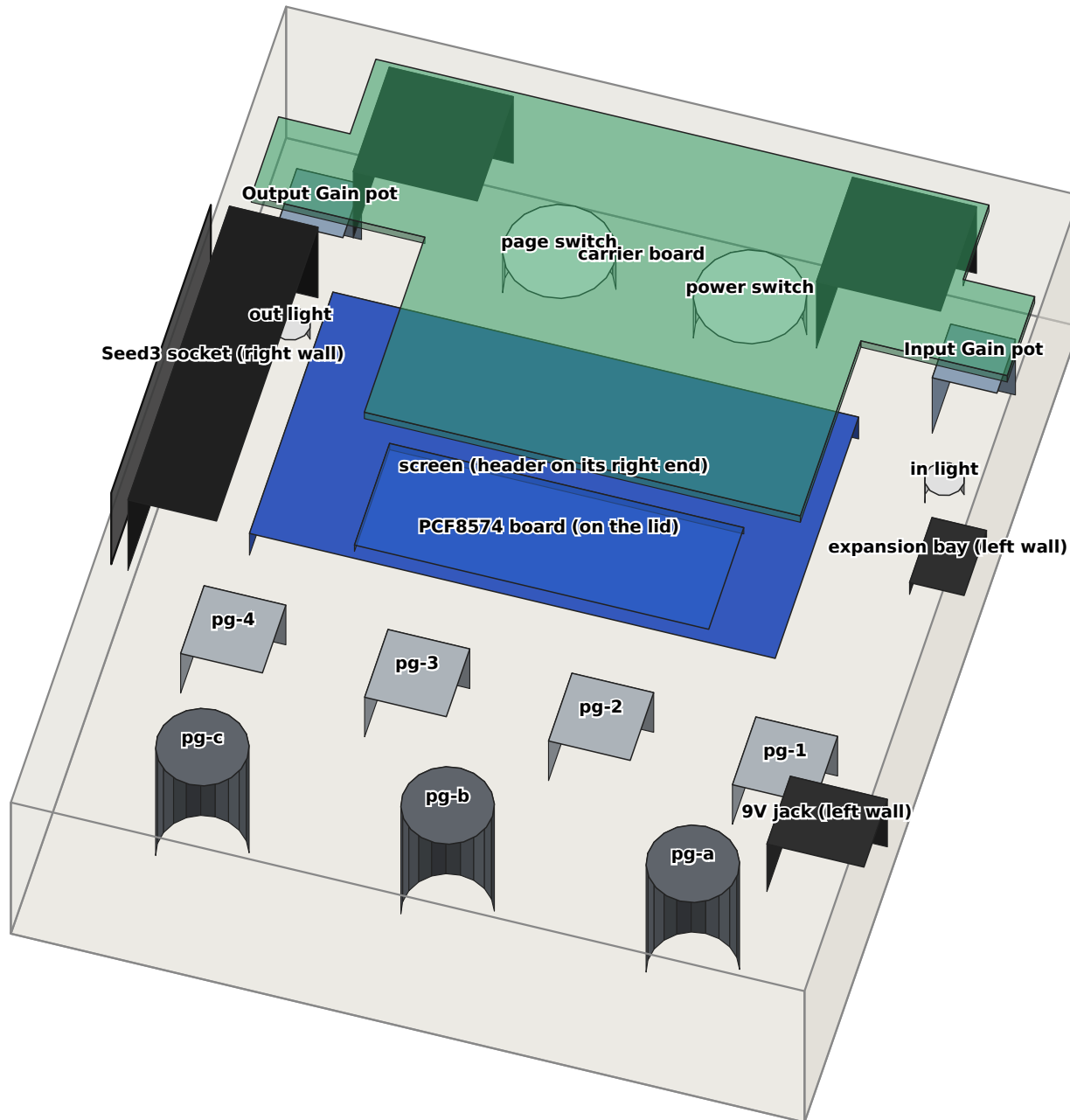


PG-1 wiring in 3D - 1. Where every part sits

Open box from above: lid off, face down on the table, the jacks' wall away from you. Seen from inside, so left and right are swapped: the in jack is on your RIGHT.



Top wall (away from you): the in / out jacks;
the carrier board hangs from their nuts,
flat, 27 mm down from the face.

Left wall (your right): 9V jack, expansion bay.

Right wall (your left): the Seed3 socket;
the Seed3 plugs in from outside.

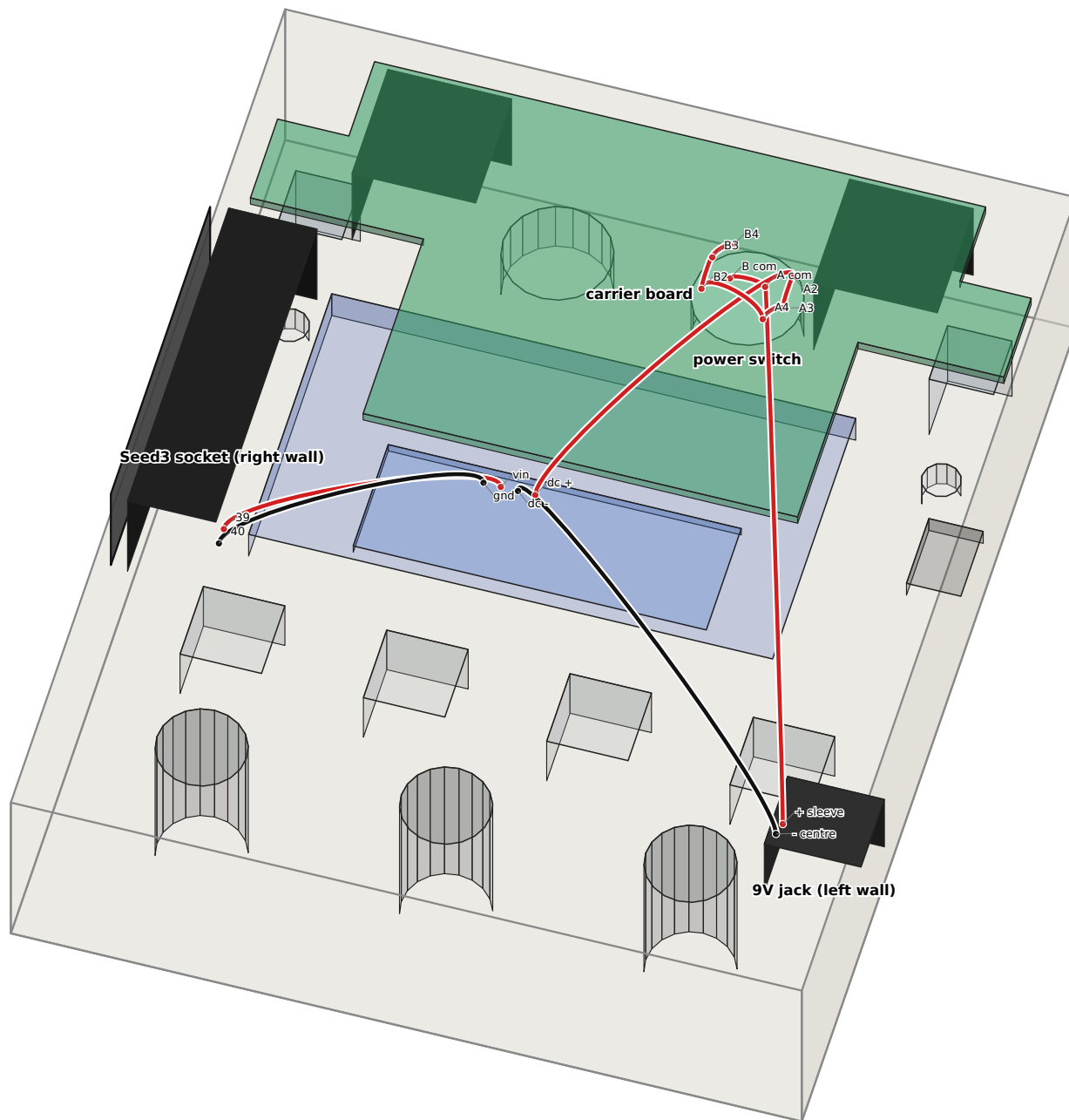
PCF8574 board: velcro on the lid, drawn
where it sits when the lid is on.

Pages 2-7: each job's wires, with the pin
names at both ends and a list in words.
Flat version with every lug: wiring-diagram.pdf

Not drawn: the 4 encoder knobs, gain knobs, chicken heads and footswitch caps (they're outside, on the face).

PG-1 wiring in 3D - 2. Power: 9 V jack -> power switch -> board -> Seed3

Open box from above: lid off, face down on the table, the jacks' wall away from you. Seen from inside, so left and right are swapped: the in jack is on your RIGHT.



Each wire, from -> to:

- 9V DC jack: + sleeve -> power switch: A com
9 V + into the switch (solder)
- power switch: A com -> power switch: B com
- power switch: A2 -> carrier J10: dc +
ON lugs -> dc + (join A2-A4 and B2-B4 with bare offsets first)
- power switch: A2 -> power switch: A3
- power switch: A3 -> power switch: A4
- power switch: A4 -> power switch: B2
- power switch: B2 -> power switch: B3
- power switch: B3 -> power switch: B4
- 9V DC jack: - centre -> carrier J10: dc -
9 V - straight to the board (female end on the header)
- carrier J10: vin -> Seed3 socket place 39
jumper: Seed3 power (9 V, protected)
- carrier J10: gnd -> Seed3 socket place 40
jumper: Seed3 ground (the ground chain is spliced onto this wire)

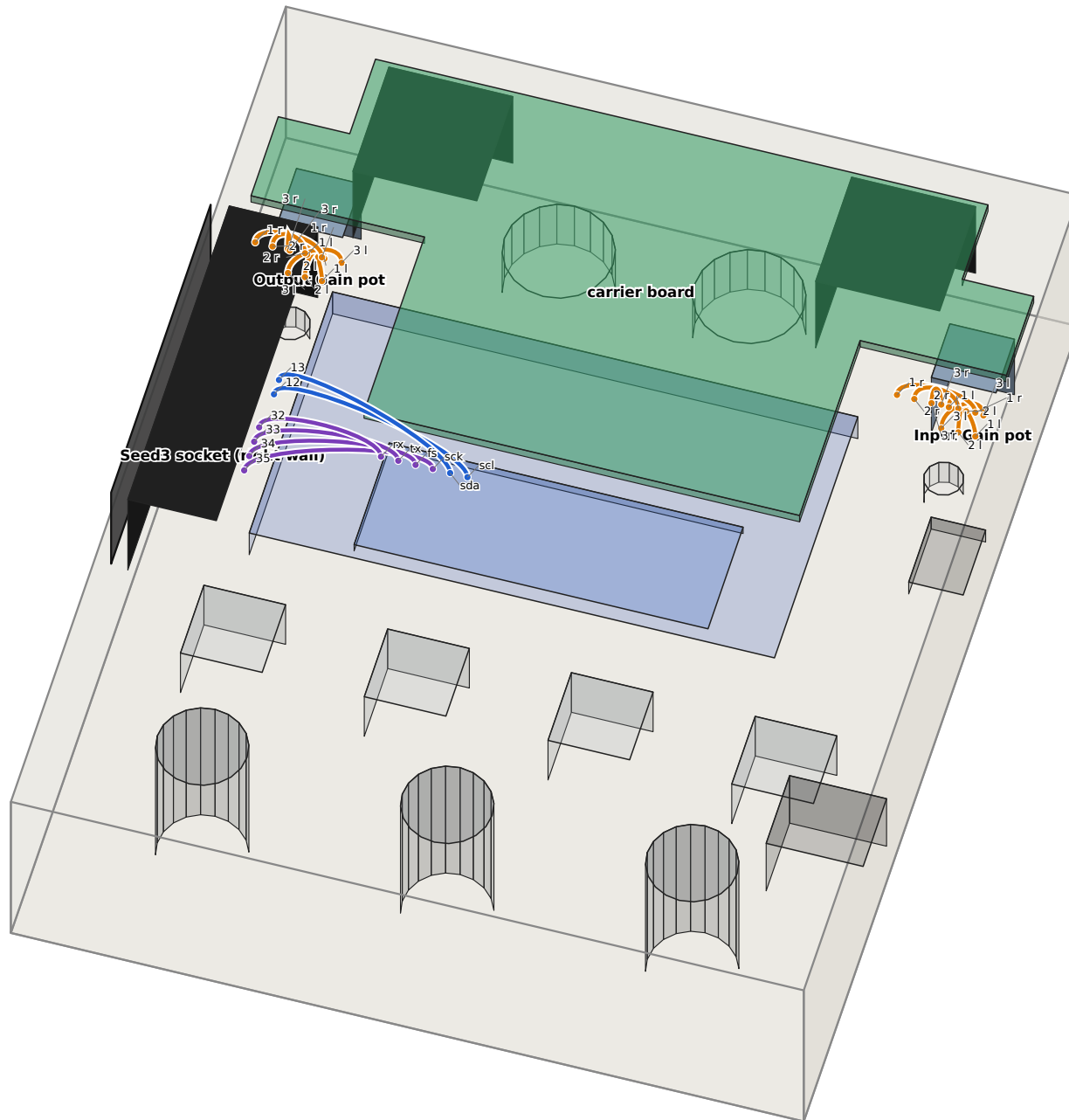
Seed3 socket places on this page (from inside):

39: vin (carrier J10)

40: gnd (carrier J10)

PG-1 wiring in 3D - 3. Board <-> Seed3, and the gain pots <-> the board

Open box from above: lid off, face down on the table, the jacks' wall away from you. Seen from inside, so left and right are swapped: the in jack is on your RIGHT.



Each wire, from -> to:

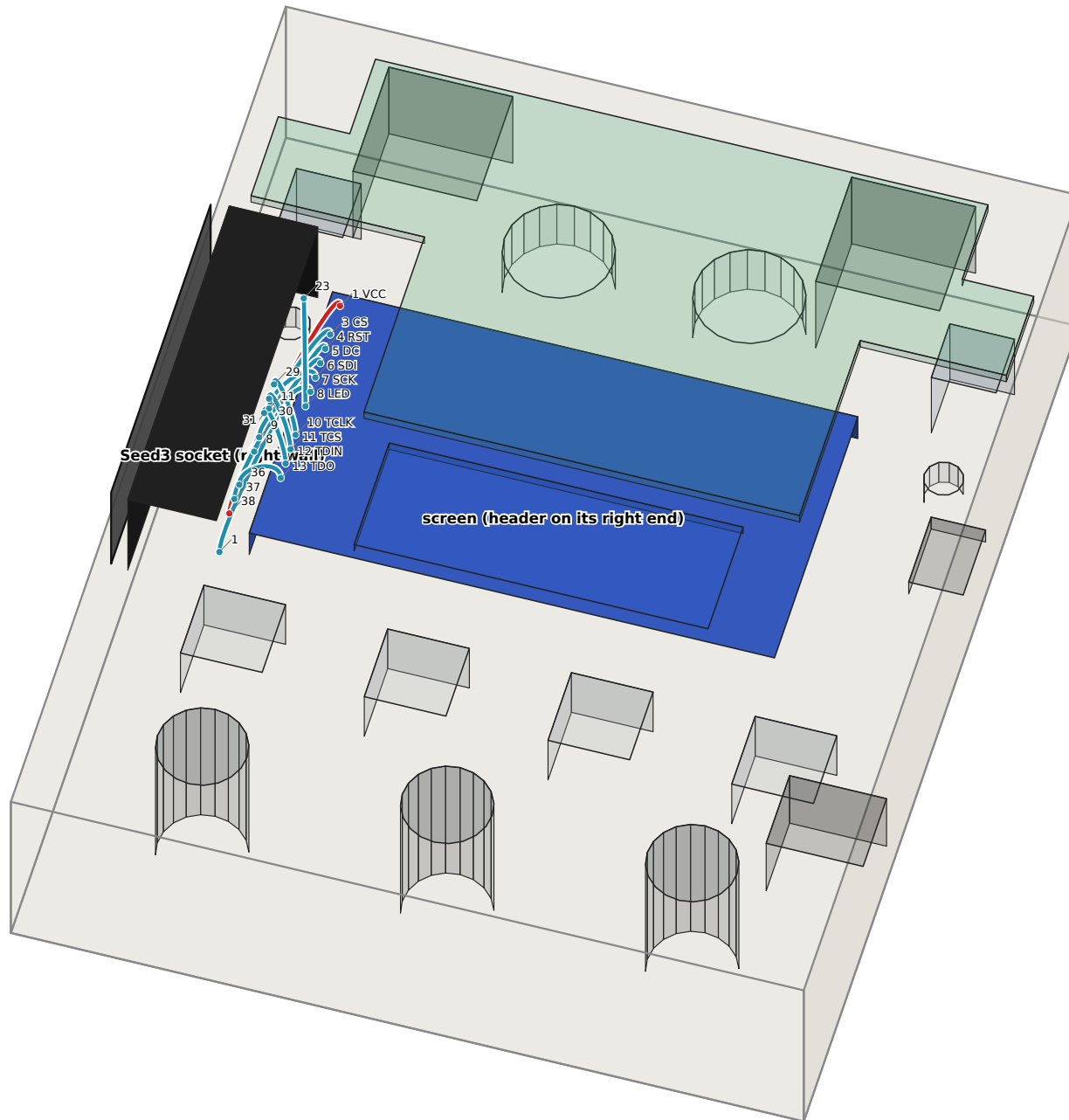
- carrier J10: scl -> Seed3 socket place 12
- jumper: I2C clock
- carrier J10: sda -> Seed3 socket place 13
- jumper: I2C data
- carrier J10: sck -> Seed3 socket place 35
- jumper: audio bit clock
- carrier J10: fs -> Seed3 socket place 34
- jumper: audio frame clock
- carrier J10: tx -> Seed3 socket place 33
- jumper: audio Seed3 -> board
- carrier J10: rx -> Seed3 socket place 32
- jumper: audio board -> Seed3
- Input Gain pot: 1 l -> carrier pg-line: 1 l
- Input Gain pot: 2 l -> carrier pg-line: 2 l
- Input Gain pot: 3 l -> carrier pg-line: 3 l
- Input Gain pot: 1 r -> carrier pg-line: 1 r
- Input Gain pot: 2 r -> carrier pg-line: 2 r
- Input Gain pot: 3 r -> carrier pg-line: 3 r
- Output Gain pot: 1 l -> carrier pg-hp: 2 l
- Output Gain pot: 3 l -> carrier pg-hp: 3 l
- Output Gain pot: 1 r -> carrier pg-hp: 1 r
- Output Gain pot: 2 r -> carrier pg-hp: 2 r
- Output Gain pot: 3 r -> carrier pg-hp: 3 r

Seed3 socket places on this page (from inside):

- | | |
|-----------------------|-----------------------|
| 12: scl (carrier J10) | 13: sda (carrier J10) |
| 32: rx (carrier J10) | 33: tx (carrier J10) |
| 34: fs (carrier J10) | 35: sck (carrier J10) |

PG-1 wiring in 3D - 4. Screen -> Seed3

Open box from above: lid off, face down on the table, the jacks' wall away from you. Seen from inside, so left and right are swapped: the in jack is on your RIGHT.



Each wire, from -> to:

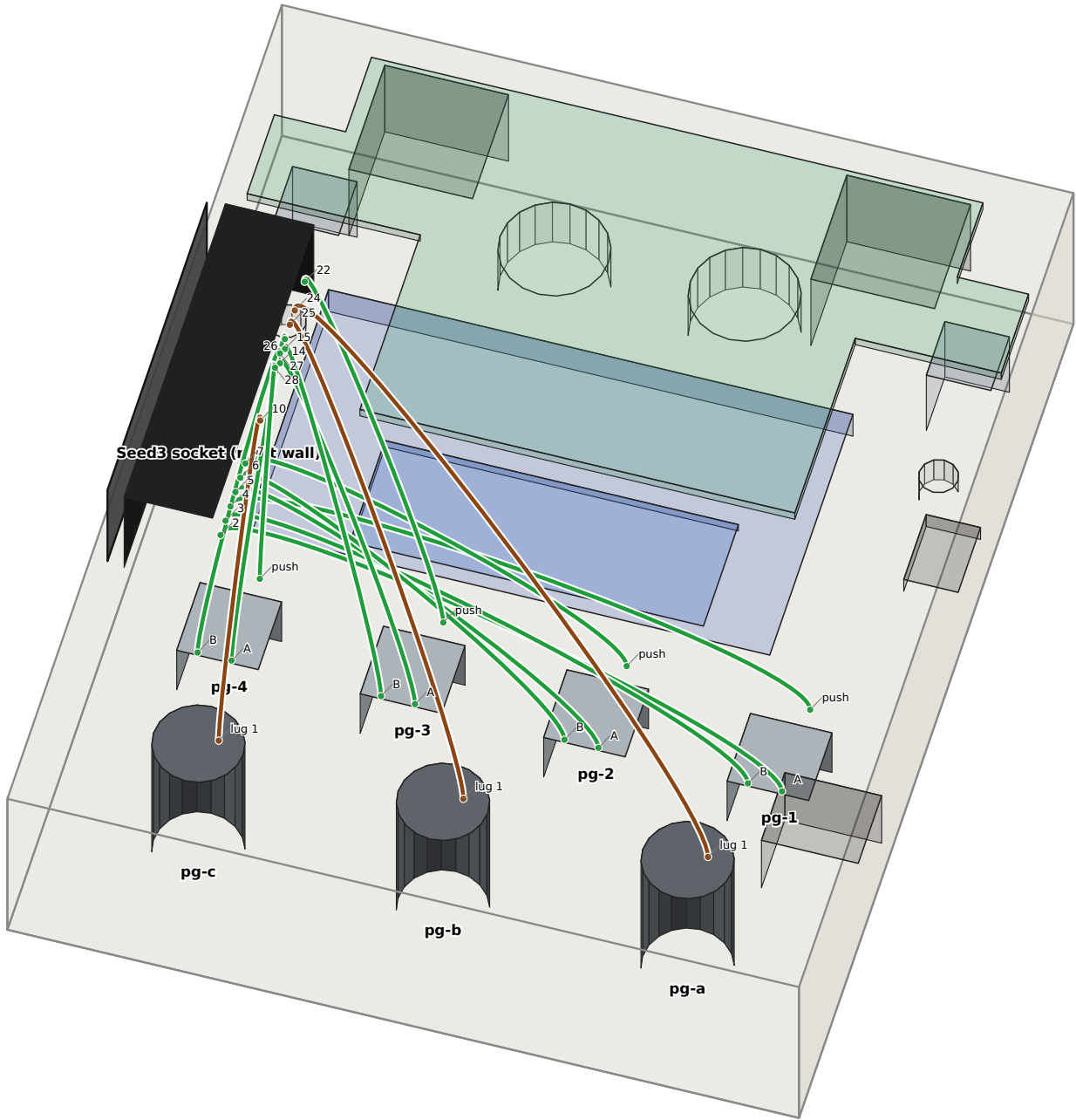
- screen A-8180: 1 VCC -> Seed3 socket place 38
jumper: 3.3 V
- screen A-8180: 3 CS -> Seed3 socket place 8
- screen A-8180: 4 RST -> Seed3 socket place 1
- screen A-8180: 5 DC -> Seed3 socket place 37
- screen A-8180: 6 SDI -> Seed3 socket place 11
- screen A-8180: 7 SCK -> Seed3 socket place 9
- screen A-8180: 8 LED -> Seed3 socket place 23
- jumper: backlight
- screen A-8180: 10 TCLK -> Seed3 socket place 29
- jumper: touch
- screen A-8180: 11 TCS -> Seed3 socket place 30
- jumper: touch
- screen A-8180: 12 TDIN -> Seed3 socket place 31
- jumper: touch
- screen A-8180: 13 TDO -> Seed3 socket place 36
- jumper: touch

Seed3 socket places on this page (from inside):

- | | |
|----------------------------|-----------------------------|
| 1: 4 RST (screen A-8180) | 8: 3 CS (screen A-8180) |
| 9: 7 SCK (screen A-8180) | 11: 6 SDI (screen A-8180) |
| 23: 8 LED (screen A-8180) | 29: 10 TCLK (screen A-8180) |
| 30: 11 TCS (screen A-8180) | 31: 12 TDIN (screen A-8180) |
| 36: 13 TDO (screen A-8180) | 37: 5 DC (screen A-8180) |
| 38: 1 VCC (screen A-8180) | |

PG-1 wiring in 3D - 5. Knobs and footswitches -> Seed3

Open box from above: lid off, face down on the table, the jacks' wall away from you. Seen from inside, so left and right are swapped: the in jack is on your RIGHT.



Each wire, from -> to:

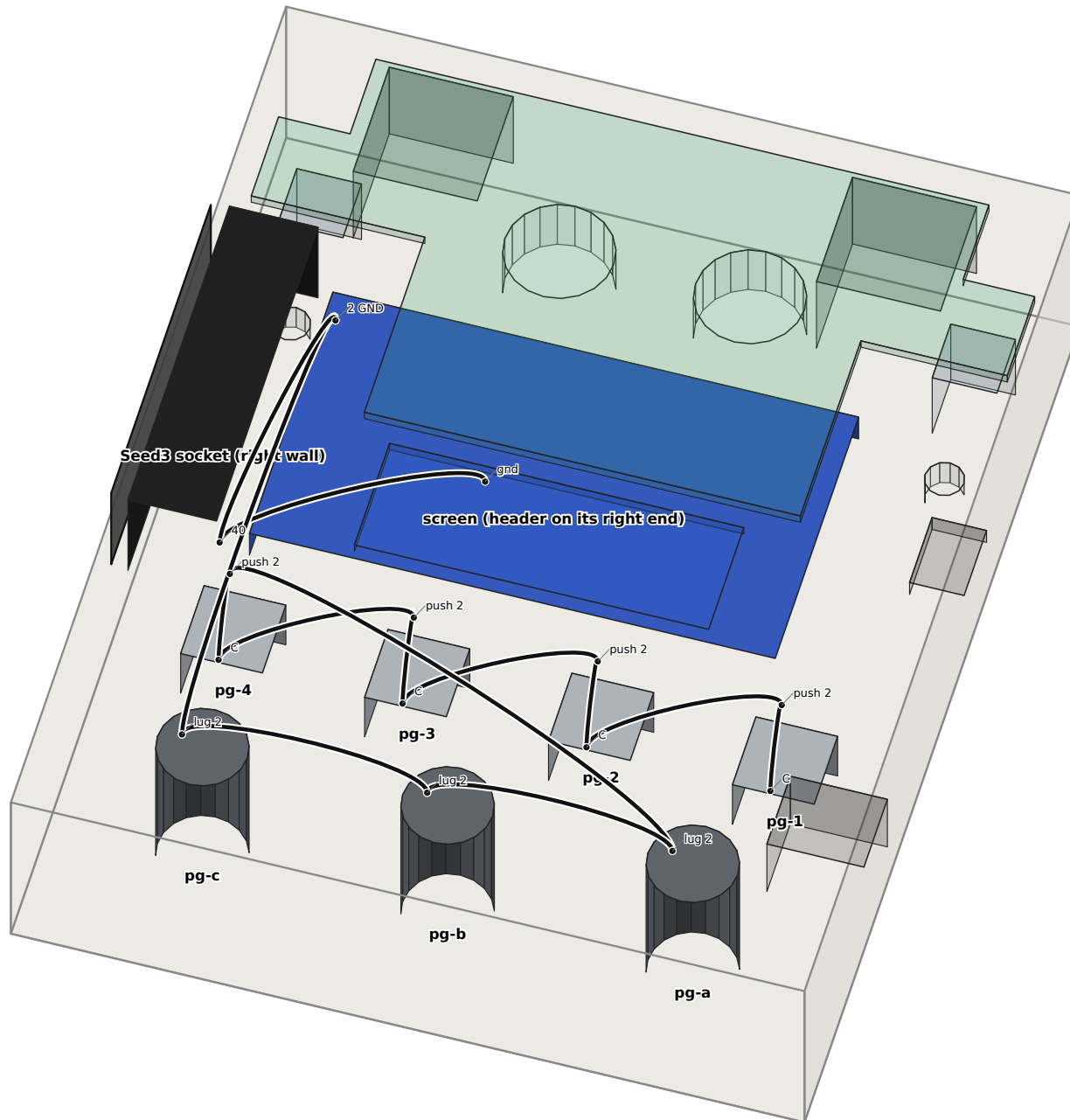
- pg-1 encoder: A -> Seed3 socket place 2
- pg-1 encoder: B -> Seed3 socket place 3
- pg-1 encoder: push -> Seed3 socket place 4
- pg-2 encoder: A -> Seed3 socket place 5
- pg-2 encoder: B -> Seed3 socket place 6
- pg-2 encoder: push -> Seed3 socket place 7
- pg-3 encoder: A -> Seed3 socket place 14
- pg-3 encoder: B -> Seed3 socket place 15
- pg-3 encoder: push -> Seed3 socket place 22
- pg-4 encoder: A -> Seed3 socket place 26
- pg-4 encoder: B -> Seed3 socket place 27
- pg-4 encoder: push -> Seed3 socket place 28
- pg-a footswitch: lug 1 -> Seed3 socket place 24
- pg-b footswitch: lug 1 -> Seed3 socket place 25
- pg-c footswitch: lug 1 -> Seed3 socket place 10

Seed3 socket places on this page (from inside):

- | | |
|-----------------------------|-----------------------------|
| 2: A (pg-1 encoder) | 3: B (pg-1 encoder) |
| 4: push (pg-1 encoder) | 5: A (pg-2 encoder) |
| 6: B (pg-2 encoder) | 7: push (pg-2 encoder) |
| 10: lug 1 (pg-c footswitch) | 14: A (pg-3 encoder) |
| 15: B (pg-3 encoder) | 22: push (pg-3 encoder) |
| 24: lug 1 (pg-a footswitch) | 25: lug 1 (pg-b footswitch) |
| 26: A (pg-4 encoder) | 27: B (pg-4 encoder) |
| 28: push (pg-4 encoder) | |

PG-1 wiring in 3D - 6. The ground chain (black wire, part to part)

Open box from above: lid off, face down on the table, the jacks' wall away from you. Seen from inside, so left and right are swapped: the in jack is on your RIGHT.



Each wire, from -> to:

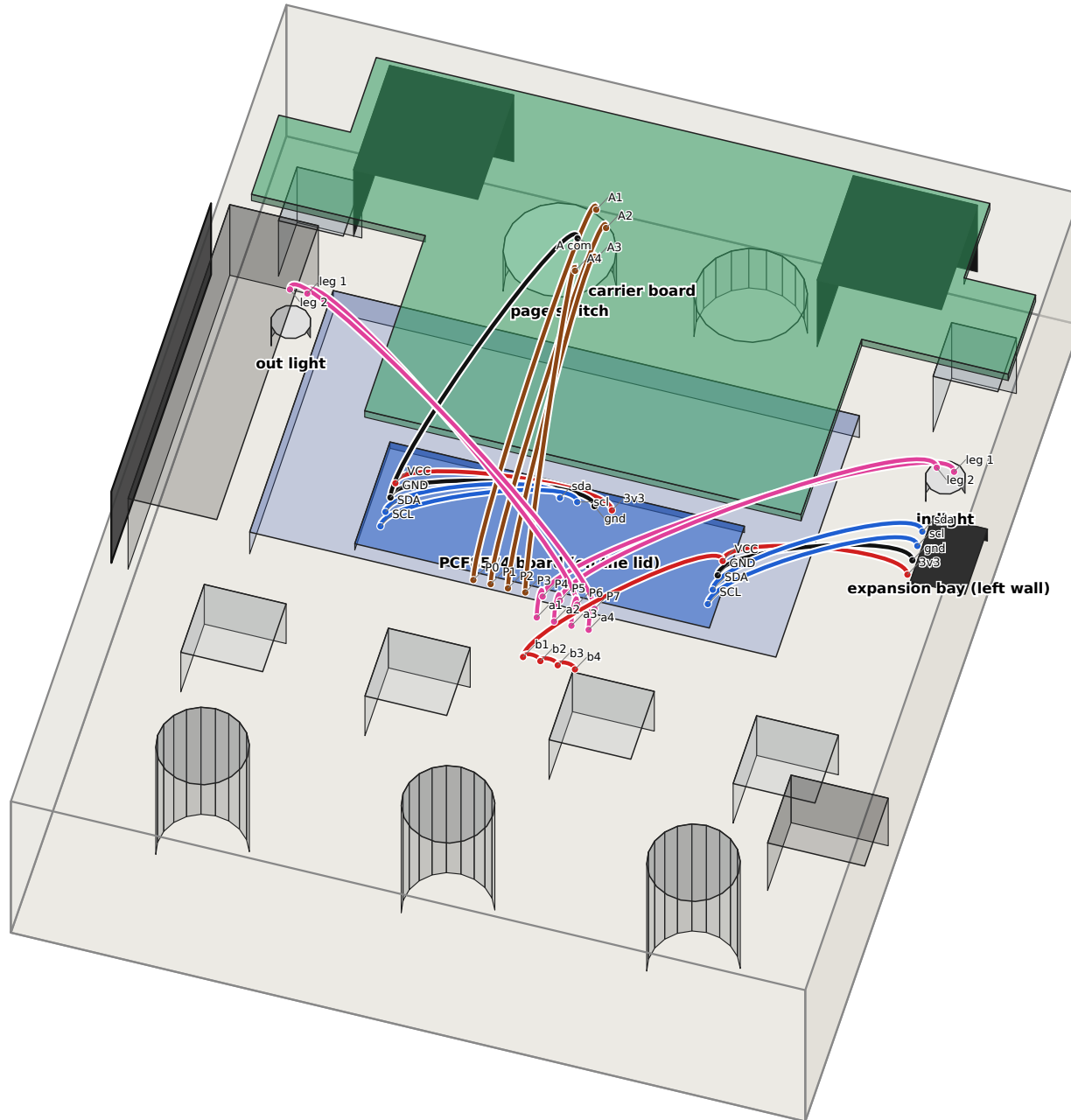
- carrier j10: gnd -> Seed3 socket place 40
- jumper: Seed3 ground (the ground chain is spliced onto this wire)
- pg-1 encoder: C -> pg-1 encoder: push 2
- pg-1 encoder: push 2 -> pg-2 encoder: C
- pg-2 encoder: C -> pg-2 encoder: push 2
- pg-2 encoder: push 2 -> pg-3 encoder: C
- pg-3 encoder: C -> pg-3 encoder: push 2
- pg-3 encoder: push 2 -> pg-4 encoder: C
- pg-4 encoder: C -> pg-4 encoder: push 2
- pg-4 encoder: push 2 -> pg-a footswitch: lug 2
- pg-a footswitch: lug 2 -> pg-b footswitch: lug 2
- pg-b footswitch: lug 2 -> pg-c footswitch: lug 2
- pg-c footswitch: lug 2 -> screen A-8180: 2 GND
- ground chain (screen GND: cut a jumper, solder its cut end in the chain)
- screen A-8180: 2 GND -> Seed3 socket place 40
- chain's end: SPLICED onto the j10.gnd -> seed.40 jumper, 3 cm from the socket (strip 5 mm, wrap

Seed3 socket places on this page (from inside):

40: 2 GND (screen A-8180)

PG-1 wiring in 3D - 7. PCF8574 board (on the lid): page switch, level lights, board, bay

Open box from above: lid off, face down on the table, the jacks' wall away from you. Seen from inside, so left and right are swapped: the in jack is on your RIGHT.



Each wire, from -> to:

- PCF8574 board: VCC -> carrier expansion: 3v3 jumper (3.3 V)
- PCF8574 board: GND -> carrier expansion: gnd
- PCF8574 board: SDA -> carrier expansion: sda
- PCF8574 board: SCL -> carrier expansion: scl
- PCF8574 board: VCC -> expansion bay: 3v3 jumper, its far end glued in the bay
- PCF8574 board: GND -> expansion bay: gnd jumper, its far end glued in the bay
- PCF8574 board: SDA -> expansion bay: sda jumper, its far end glued in the bay
- PCF8574 board: SCL -> expansion bay: scl jumper, its far end glued in the bay
- page switch: A1 -> PCF8574 board: P0 position 1 (find with beep mode)
- page switch: A2 -> PCF8574 board: P1 position 2
- page switch: A3 -> PCF8574 board: P2 position 3
- page switch: A4 -> PCF8574 board: P3 position 4
- page switch: A com -> PCF8574 board: GND pole A common: solder it on the UNDERSIDE of the PCF board, on the GND pin's solder joint
- in level light: leg 1 -> PCF8574 board: P4 in light (swap its 2 legs, or set kSwapLightColours, if the colours are backwards)
- in level light: leg 2 -> PCF8574 board: P5 in light
- out level light: leg 1 -> PCF8574 board: P6 out light
- out level light: leg 2 -> PCF8574 board: P7 out light
- 4 x 330 ohm: a1 -> PCF8574 board: P4 330 ohm pull-up: solder its leg on the PCF board's underside, on P4's joint
- 4 x 330 ohm: a2 -> PCF8574 board: P5 330 ohm pull-up
- 4 x 330 ohm: a3 -> PCF8574 board: P6 330 ohm pull-up
- 4 x 330 ohm: a4 -> PCF8574 board: P7 330 ohm pull-up
- 4 x 330 ohm: b1 -> PCF8574 board: VCC the 4 resistors' other ends joined -> underside of a VCC pin's joint
- 4 x 330 ohm: b1 -> 4 x 330 ohm: b2
- 4 x 330 ohm: b2 -> 4 x 330 ohm: b3
- 4 x 330 ohm: b3 -> 4 x 330 ohm: b4